Predicting Domain-Insertion Tolerance in SpCas9 with a Bayesian Additive Regression Trees (BART) Model

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1. Problem and framing

Inserting one folded protein domain into another is a common protein-engineering strategy, but most positions do not tolerate it, since an insertion at an unsuitable site causes the host to misfold or lose activity. The objective is to predict which positions of SpCas9, a 1,368-residue protein, tolerate an inserted folded domain. I frame the problem as one of ranking and triage rather than of classification accuracy, because the intended consumer is a wet-lab campaign that can test only a small number of sites and requires the most promising candidates first. This framing motivates two modelling choices defended below: a calibrated probability per residue, and an accompanying per-site uncertainty.

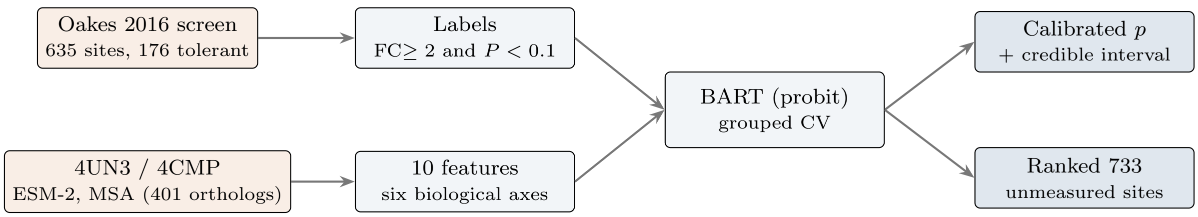

Figure 1: Overview of the pipeline. A public functional screen provides the labels; structural and sequence sources provide the features; a single BART classifier is evaluated under grouped cross-validation and then applied to rank the 733 residues the screen did not measure.

2. Data and its caveats

The labels are derived from the Oakes et al. (2016) screen. The authors inserted an 86-residue PDZ domain across Cas9, subjected the library to a CRISPRi functional selection using catalytically dead Cas9, and sequenced the pool before and after selection. Functional variants survive and become enriched, whereas non-functional variants are depleted. Each surviving position is reported with a fold change, the ratio of its post-selection to its pre-selection abundance, and an associated P-value.

Parsing the data required several deliberate decisions. The supplement contains only significant entries (636 rows, maximum P of 0.0996), so the $P < 0.1$ threshold is the authors', not mine. I removed the single row at `AA=0`, an insertion N-terminal to residue 1 with no residue to map, which leaves 635 measured sites. Two clones are absent from the pre-screen library and report infinite fold change; I treat them as strongly enriched. Site `AA=i` is mapped to the features of residue i, the transposon's short duplication having already been collapsed into SpCas9 numbering by the authors. The principal concern is coverage: the screen reached only about 70% of positions, and the significant table is a subset of those, so 733 residues are unlabelled. An unmeasured residue is not an intolerant one, and labelling these 733 as negatives would fabricate data and degrade calibration. They are therefore withheld entirely and constitute the prediction set that the model scores and the laboratory consumes.

3. Target: binary, with a defended threshold

I label a site tolerant when its fold change is at least 2 and its P-value is below 0.1. The target is binary by design: the decision it supports is itself binary, namely whether or not to attempt an insertion, and a calibrated probability is what allows a laboratory to rank and threshold candidates. The two-fold cut is the conventional definition of meaningful enrichment; it yields 176 positives out of 635, a base rate near 28%. With ten features this corresponds to roughly ten events per variable, a ratio that limits overfitting. A regression target on `log2` fold change remains available as an alternative. It retains more information, but its tails are poorly behaved (fold changes of zero and of infinity), and the operational question is binary, so the binary target is primary.

4. Features and why each should carry signal

I selected ten features to characterise the environment of each residue, organised into six biological axes; the grouping supports the rationale and importance reporting, while the model receives only a flat ten-dimensional vector. Three features address whether there is physical room for an insertion: relative solvent-accessible surface area, local atomic packing within 18 Å, and the crystallographic B-factor. Two address disruption of nucleic-acid binding, the minimum distances to the guide RNA and to the DNA, since the binding channels are selected against. One addresses local rigidity, the distance to the nearest helix or strand terminus, which separates tolerant element edges from rigid mid-element positions. Three address evolutionary constraint: ESM-2 entropy, alignment conservation, and indel frequency, since positions that tolerate natural indels tend to tolerate engineered insertions. The final axis addresses conformational dynamics.

The conformational-dynamics feature was added last and is the most mechanistically motivated. SpCas9 undergoes a large conformational change upon activation, so an insertion in a region that must move during that transition tends to abolish function. I quantify this by superposing the apo (4CMP) and holo (4UN3) structures on the rigid RuvC core (261 residues, 2.37 Å RMSD) and measuring the per-residue Cα displacement between the two states. The separation between classes is pronounced: tolerant sites move 14.6 Å on average, intolerant sites 26.9 Å. Conservation is computed by two independent methods, the ESM-2 protein language model and a MAFFT alignment of 401 orthologs, so that the resulting signal does not depend on either alone.

5. Model

I train a single model: a Bayesian Additive Regression Trees (BART) classifier with a probit link, fitted with `dbarts`. The choice suits the data. With only 176 positives, a self-regularising model is preferable to one requiring extensive tuning, and BART regularises through priors on tree depth and leaf magnitude. As a sum of shallow trees, it also captures nonlinearities and interactions without manual specification, which is relevant because several features are non-monotonic; tolerated positions tend to lie a short distance from the DNA rather than maximally far from it. Most importantly for this application, BART is Bayesian, so 1,000 posterior draws yield both a calibrated probability and a 95% credible interval per residue.

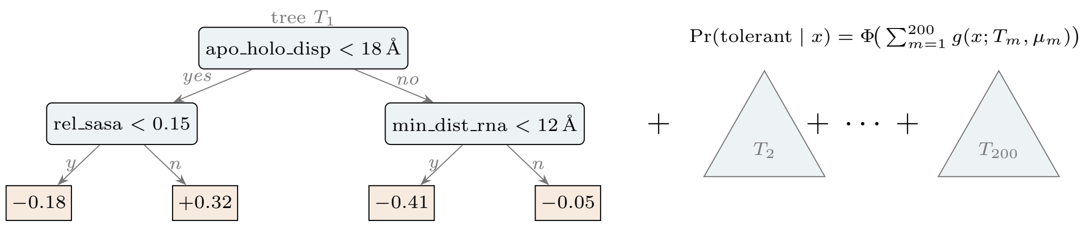

Figure 2: Structure of the BART model. The latent score for a residue is the sum of 200 shallow trees, mapped through the probit link Φ to a probability. The example tree T_1 splits first on conformational displacement, then on exposure or proximity to the guide RNA; positive leaf values increase the predicted tolerance. Posterior draws over the ensemble yield the per-residue credible interval.

The depth prior restricts trees to two or three levels and the leaf prior ($k = 2$) shrinks their contributions, so at this sample size the regularisation dominates and the ensemble does not overfit. I verified this rather than assuming it. A small grid over the leaf-shrinkage and tree-count parameters changed the honest out-of-fold score by at most 0.02, and the optimism gap, the difference between the best inner-loop score and the honest out-of-fold score, was only 0.073. A gap this small indicates that further search would fit cross-validation noise rather than signal, so the defaults were retained ($k = 2$, `ntree`= 200, 1,000 draws).

6. Honest evaluation

Residues are spatially autocorrelated, since positions adjacent along the chain resemble one another, so a random k-fold split leaks neighbouring residues across the train-test boundary and inflates apparent performance. I therefore use two grouped splits that bracket the deployment regime. Block cross-validation partitions the sequence into eight contiguous segments, approximating the setting in which scored residues are interspersed among measured ones. Leave-a-domain-out cross-validation withholds an entire Cas9 domain at a time, a stricter test of generalisation to unseen structure. AUPRC is the primary metric, because the classes are imbalanced and it reflects the quality of the ranked shortlist the laboratory acts upon; precision in the top 20 and top 50 is reported for the same reason. Accuracy is uninformative here, as predicting "intolerant" for every residue already attains 72%.

Table 1: Out-of-fold performance of the final BART model (ten features), base rate 0.28. The two schemes bracket deployment, and their close agreement says the model is not leaning on shortcuts tied to which domain a residue sits in.

Cross-validation scheme	n_{OOF}	AUPRC	AUROC	P@20	P@50	Brier
Leave-a-domain-out (strict)	627	**0.637**	0.809	0.70	**0.82**	0.149
Contiguous block	635	0.645	0.837	–	–	–

P@k and Brier are for the headline (domain) run; block is shown for the robustness comparison.

The progression toward these results is informative. The headline AUPRC increased from 0.50 to 0.59 to 0.64 across three rounds of feature work, none of it attributable to tuning (Figure 3). The first gain followed the removal of a leaking feature; the second followed the removal of two redundant features and the addition of the dynamics feature. Against a base rate of 0.28, the final model places 41 of its 50 highest-ranked sites on genuinely tolerant residues.

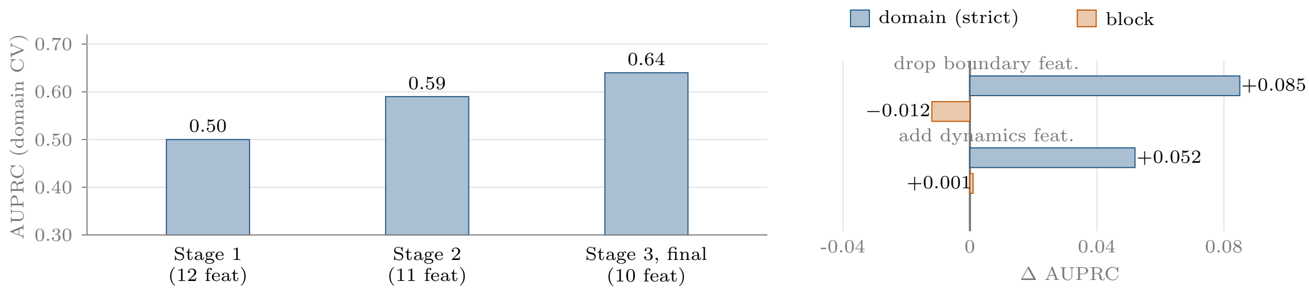

Figure 3: Left: headline AUPRC increases with feature work, not with tuning. Right: the leakage test. Dropping `dist_to_domain_boundary` improves the strict split (+0.085) but not the block split (−0.012), the signature of a feature that implicitly encodes domain identity. The dynamics feature shows the opposite pattern, improving the strict split.

7. What the model learned, and a detected leakage

Aggregated by axis, BART's variable inclusion ranks exposure first (≈ 0.30), followed by proximity to the nucleic acids (≈ 0.20), conservation (≈ 0.20), and dynamics (≈ 0.10). This ordering is consistent with the biology: a favourable insertion site has available room, lies away from the binding channels, is not part of a conserved core, and is not required to move during activation. The leakage test corrected an initial assumption. The feature `dist_to_domain_boundary`, added on biological grounds, implicitly encodes the domain to which a residue belongs. Removing it changed the block score negligibly (−0.012) but raised the strict leave-a-domain-out score by 0.085 (Figure 3, right). This asymmetry, a benefit when the domain is represented in training and a penalty when it is not, is the characteristic signature of leakage, and the feature was removed. The dynamics feature exhibits the opposite pattern and was retained. Ablation determined the remaining selections (Table 2): univariate strength does not imply marginal value, so features were retained by their marginal contribution rather than by their performance in isolation.

Table 2: Feature ablation under leave-a-domain-out CV (gradient-boosting proxy). Production features baseline AUPRC 0.558. Positive drop-one and add-one values mean the feature helps. The retained dynamics feature is marked; several mechanism-driven candidates were rejected.

Feature	Group	Univar.	Drop1	Feature	Group	Univar.	Add1
min_dist_rna	prod	0.545	+0.003	solvent_cone	novel	0.502	+0.001
flexibility	prod	0.516	+0.002	min_dist_heterodup.	novel	0.482	−0.006
open_volume_18A	prod	0.455	−0.006	contact_order	novel	0.452	+0.001
rel_sasa	prod	0.448	−0.008	**apo_holo_disp**	**novel**	0.394	**+0.018**
indel_frequency	prod	0.363	+0.002	long_range_contacts	novel	0.379	−0.013
dist_to_sse_end	prod	0.347	−0.008	min_dist_na_ends	novel	0.357	+0.009
msa_conservation	prod	0.319	**+0.038**	min_dist_nontarget	novel	0.344	−0.019
esm2_entropy	prod	0.295	−0.010	true_insertion_freq	novel	0.237	−0.020

A normal-mode elastic-network model was tested as an alternative dynamics descriptor: using ProDy, I built an Anisotropic Network Model on the holo Cα trace and took each residue's fluctuation in the ten slowest collective modes (`anm_msf`). Although it covers every resolved residue, it did not help, lowering AUPRC slightly under both schemes (0.635 to 0.621 under leave-a-domain-out, 0.645 to 0.639 under block) and performing clearly worse as a substitute for `apo_holo_disp` (0.575 under leave-a-domain-out). The model reports the intrinsic equilibrium fluctuations of a single wild-type structure, a smooth profile dominated by termini and surface loops, so it overlaps the existing B-factor feature and is far less specific to the apo-to-holo activation motion that `apo_holo_disp` captures directly from two distinct states.

Two further failure modes should be noted. The non-monotonic dependence on distance to DNA is only partially captured, and a decomposition into specific channel distances did not improve on the broad feature.

8. Limitations

The metrics are optimistic by construction, since training and evaluation are restricted to significant sites; the 0.82 top-50 precision is a ceiling on a clean subset rather than a deployment guarantee, and only experimental validation can establish the true operating point. Confidence intervals on the deltas were not computed, and with six folds and 176 positives a difference such as 0.64 versus 0.59 is partly noise. The dynamics feature is roughly one-third imputed (residues unresolved in the apo structure), with non-random missingness, so part of its value may be artefactual. RuvC is discontinuous in sequence, so only HNH, PI, and REC are genuinely unseen-domain holdouts. Finally, this concerns a single inserted domain in a single host protein scored against static crystal structures, so generalisation to other inserts is untested.

The first limitation is the most important: the model is evaluated on the 635 measured sites but deployed on the 733 unmeasured ones, so I quantified the gap between them (`covariate_shift.py`, Table 3). The shift is moderate and concentrated, not pervasive. A gradient-boosted classifier separates the two sets with AUC 0.68, and 13% of the deployment set falls outside the measured 1st–99th percentile envelope and is thus scored by extrapolation. It is dominated by the apo-to-holo displacement, which is both the most shifted and the most imputed feature, so its contribution to the product scores is the least trustworthy; the sequence features barely move. Because tree ensembles extrapolate flat, BART's credible intervals reflect posterior uncertainty given the training support rather than out-of-support uncertainty on that tail; the principled remedy is conformal prediction with a support check.

Table 3: Covariate shift between the evaluation set (635 measured) and the deployment set (733 unmeasured), per feature, sorted by Kolmogorov–Smirnov statistic. Standardized mean difference is (unmeasured − measured) / pooled SD. Overall: classifier separability AUC 0.68; 13% of the deployment set lies outside the measured 1st–99th percentile envelope.

Feature	Measured med.	Unmeasured med.	Std. mean diff.	KS
apo_holo_disp	3.67	28.39	+0.285	**0.247**
flexibility	37.99	42.39	+0.179	0.107
open_volume_18A	617	599	−0.150	0.097
min_dist_rna	14.54	12.92	−0.165	0.087
indel_frequency	0.140	0.234	+0.095	0.077
msa_conservation	0.318	0.327	+0.138	0.072
min_dist_dna	16.84	17.08	+0.061	0.053
rel_sasa	0.167	0.183	+0.022	0.045
esm2_entropy	1.110	1.120	+0.013	0.038
dist_to_sse_end	0.00	0.00	+0.020	0.018

9. What I would do next

With additional time, the highest-value addition is a feature that directly simulates the insertion: folding local Cas9-PDZ chimera windows with ESMFold and reading the predicted confidence (pLDDT) over the Cas9 flank, the one descriptor that models the grafted domain rather than its environment. I would also add bootstrap confidence intervals to the feature deltas. With more data, the next step is prospective validation: testing the highest-ranked sites at the bench, recalibrating on the results, and adding a `log2`-fold-change regression head for finer ranking.